TIME SERIES ANALYSIS OF VOLCANIC TEMPERATURE MONITORING: AN EXAMPLE FROM THE NISYROS ACTIVE VOLCANO AT THE HELLENIC VOLCANIC ARC

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Abstract

An efficient mitigation of a volcanic natural hazard is the continuous monitoring and separation from interference. In this research, we examine the results of a decomposition analysis from a caldera's surface temperature data and associated climatic variables. Nisyros active volcano located at the eastern segment of the recent Hellenic Volcanic Arc, and it has shown temperature surface oscillation that may reflect repeated heat emission from a currently active hydrothermal system that mostly is affected by the seasonal meteorological parameters and the atmospheric air. Time series from the meteorological station of Kos island have been utilized, in order to investigate the atmospheric influence to the time series of the surface temperature fluctuations of Nisyros caldera. The volcanic temperature raw data have been obtained from a temperature TinyTag Plus probe TGP-4520 that has been established at the southern part of the caldera nearby an active fumarolic emanation area. Temperature surface readings from August of 2016 to January of 2017 were analyzed and decomposed into three different components which are long-, seasonal- and short-term, respectively. For this purpose, both the Kolmogorov-Zurbenko and the Autoregressive Distributed Lag (ARDL) model have been incorporated. In comparison with the multiple linear regression (MLR) model, the application of the time series regression (TSR) model and the ARDL model, provided the opportunity to integrate lag operation from the dependent and the independent variables. Thus, both the coefficient determination (R^2) of the three components (long-, seasonal- and short-term) increased and the physical interpretation of the lagged values strengthened the accuracy of the predicted model. The total explanation of the predicted model was 75% and the unexplained part was 25%, whereas the unexplained part of the raw data was 66%. Therefore, the utilized methodology has increased the overall explanation of the model's forecast by more than three times. The decomposition of the temperature readings and isolating the atmospheric factor may allow an in-depth understanding of the heat circulation and emission of the hydrothermal system underneath the volcanic caldera surface.

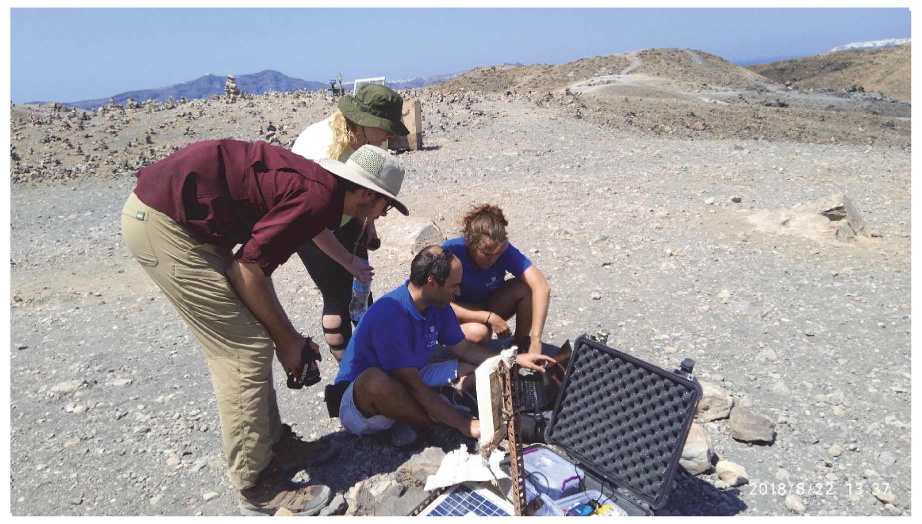

Figure 1: A volcanic monitoring station on a fumarole which is located at an active volcano (Thera) at the Hellenic Volcanic Arc in South Aegean Sea, Greece.

Introduction

Monitoring natural hazards such as volcanic eruptions, floods or earthquakes utilizing time series, is one of the most remarkable and rapidly developing techniques for prediction of disastrous events [1,2,6]. The volcano-hydrothermal system of Nisyros constitutes the first region at the Hellenic Volcanic Arc (HVA), into which time series is applied. We consider that isolating the interferences from the employed time series, in order a more thorough outcome to be exported, can provide the scientific community with useful information about the effect of atmospheric air and meteorological factors upon the volcanism in the region of Nisyros. The accomplishment of the specific procedure has been fulfilled by the application of the KZ filter, the TSR and the ARDL models.

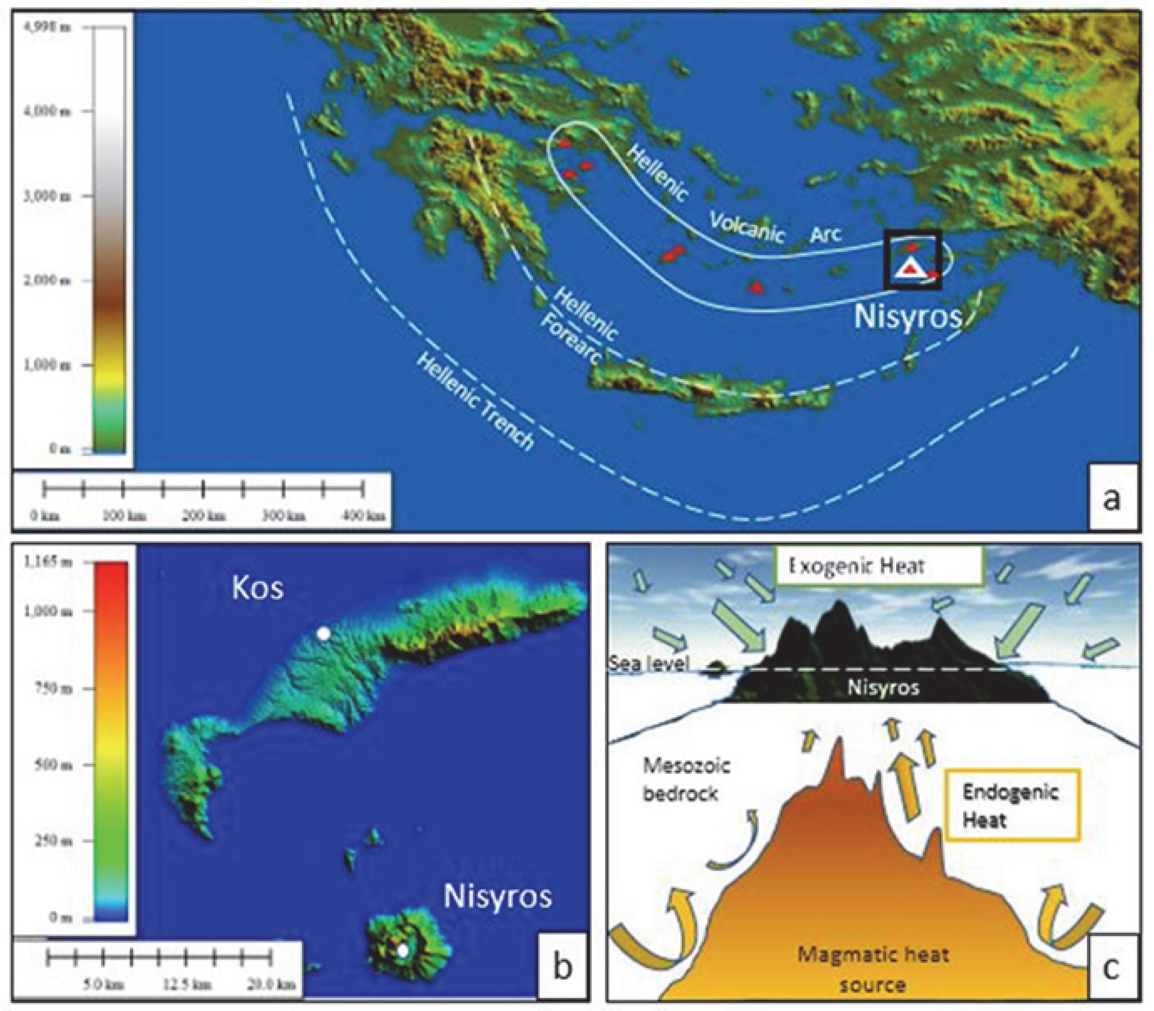

Figure 2: (a) Location of the Nisyros volcano in HVA. (b) Digital elevation model of Nisyros and Kos islands, the Kos meteorological station and the ground thermometer at Nisyros surface caldera. (c) Representation of the interaction between exogenic and endogenic heat recorded in the ground thermometer.

Methods

- Elaboration and incorporation of atmospheric air, such as temperature and pressure, and meteorological, such as humidity, wind speed, wind gust and dew point, variables into both the KNIME and SPSS software.
- Decomposition of the raw data utilizing the KZ filter into long- (LT), seasonal- (SE) and short-term components.
- Application of the TSR model in both the LT and SE components.
- Employment of the ARDL model in the ST component.

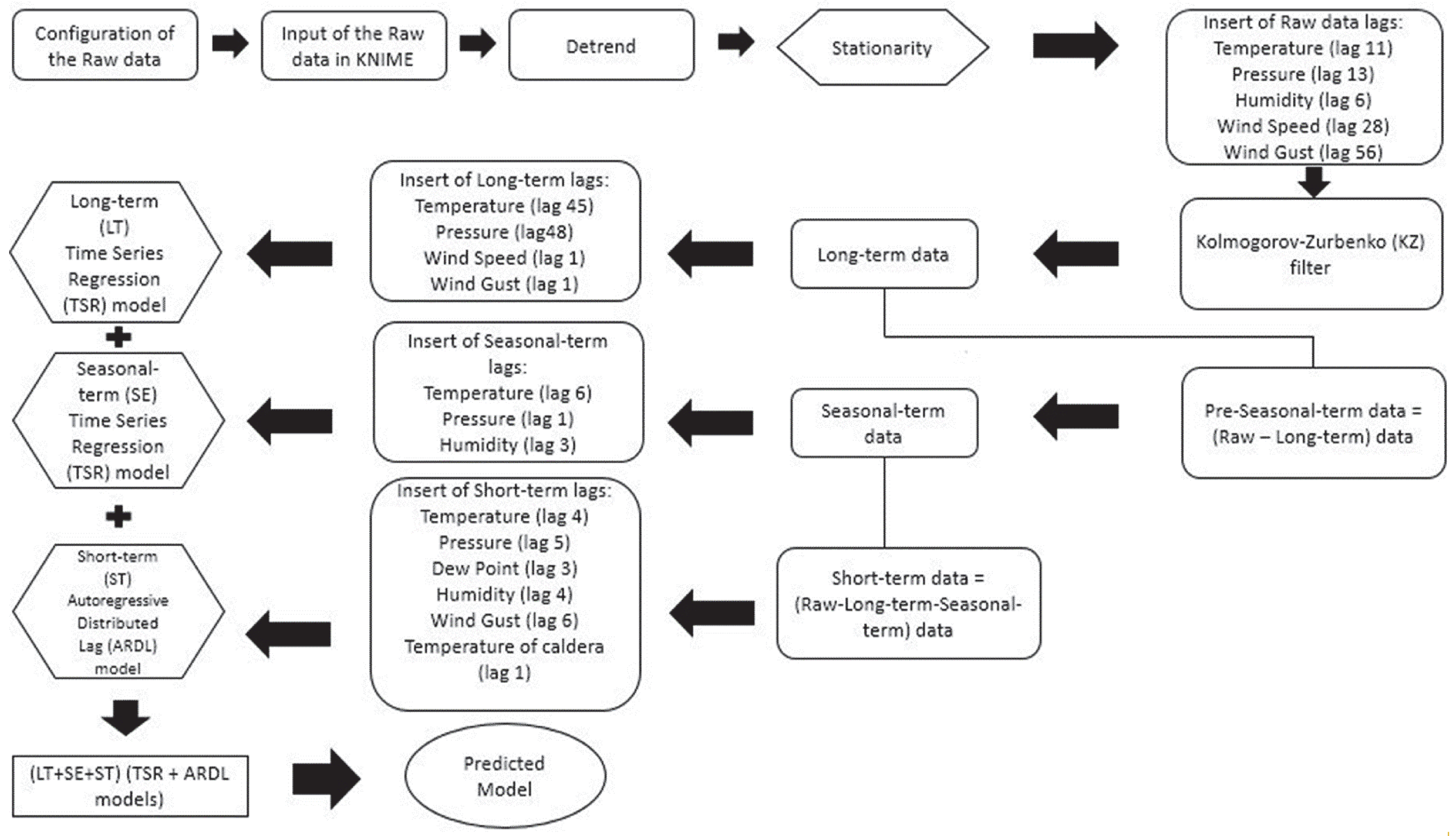

Figure 3: The applied methodology in which the KZ filter, the long and the seasonal TSR models and the short ARDL model are included.

Results

Table 1: R² of the raw, long-, seasonal- and short-term data and % variance, explanation and total explanation of the long-, seasonal- and sort-term, respectively.

	Raw data	LT	SE	ST
Coefficient Determination (R^2)	0.340	0.720	0.989	0.560
Variance %		0.190	44.090	55.720
Explanation %		0.140	43.610	31.200
Total Explanation %				74.95

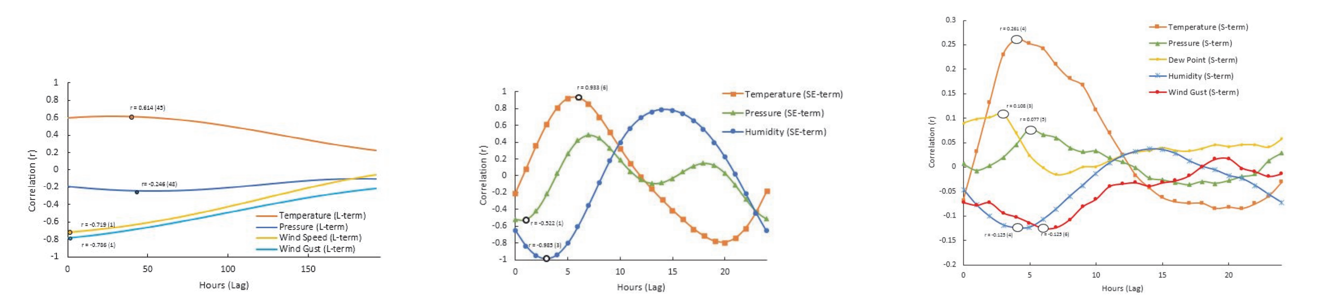

Figure 4: (Left figure) Graph of the long TSR component. (Middle figure) Line plot of the seasonal TSR component. (Right figure) Schematic representation of the short ARDL component.

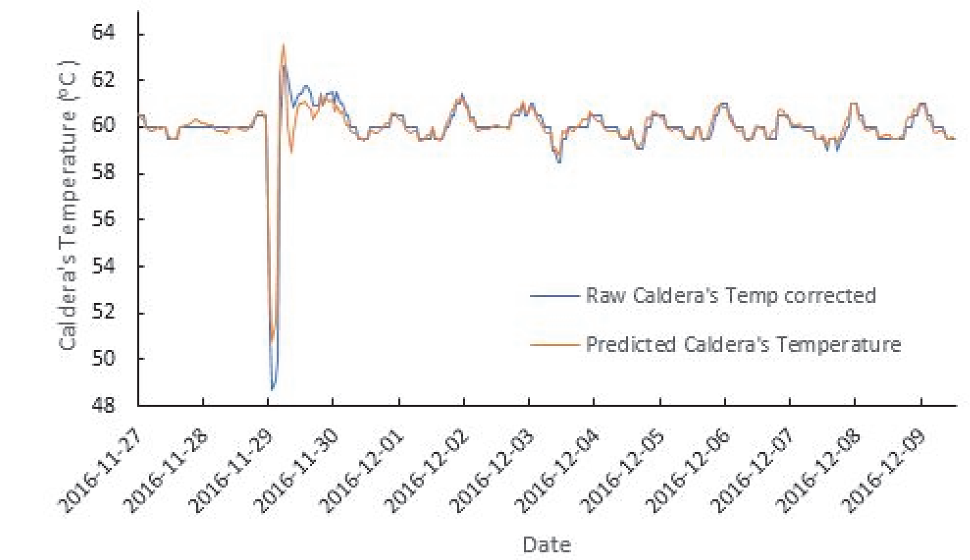

Figure 5: Line plot among the raw and the predicted caldera's temperature.

Discussion

- The total explanation of the predicted model has been augmented by the integration of lagged-values using the TSR and the ARDL models compared to the MLR model.
- The accuracy of the model's forecast has been strengthened by the physical interpretation of the lagged-values of the independent variables.

Conclusion

The utilized methodology seem to be very effective as the total explanation of the forecasting model is approximately 75% and it has been increased by more than three times compared to the raw data, whereas the incorporation of the independent variables lagged-values permitted the physical interpretation of the entire model.

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